TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Daniel Beer	Email	daniel.beer@idmt.fraunhofer.de
Institute Name	Fraunhofer Institute for Digital Media Technology IDMT			
Address	Ehrenbergstrasse 31, 98693 Ilmenau, Germany			

2. Technology Overview

Title of Proposed Technology	Ultra-Thin Flat Loudspeaker Solutions for Lightweight, Space-Saving and Highly Design-Flexible Audio Systems in Cars, Trains, Aircraft and Ships			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (**x**), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (), Next-gen Communication (**x**), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electronica • Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Imagine/Audio Devices	Speakers, Car Audio Systems		
Concept of Proposed Technology	The audio reproduction system is a key differentiator for competitive interior concepts in cars and other passenger transport vehicles. Beyond delivering a convincing listening experience, future audio systems must enable **quasi-invisible, space-saving and design-flexible loudspeaker integration** – requirements that can no longer be met with conventional cone loudspeakers. The proposed concept is a **novel flat loudspeaker technology based on arrays of micro speakers**, forming a few-millimeter thin "sounding layer" that can be mounted directly on interior surfaces such as doors, dashboards, roofs and seats (1), (2). The array geometry can closely follow the underlying surface, without being constrained to traditional circular cut-outs. As a result, the loudspeaker system can be integrated into **non-planar and curved surfaces**, for example rounded dashboards or contoured door trims, while remaining visually unobtrusive. If required, the micro speaker array can be combined with **advanced array signal processing** to create adaptive radiation characteristics (e.g. beam steering, targeted sound zones, improved speech			

	intelligibility). This enables functionalities that are difficult or impossible to achieve with today's single-driver loudspeakers. Compared to state-of-the-art conventional loudspeakers, the proposed flat loudspeaker array offers: • **Significantly reduced installation depth and weight**, supporting lightweight design and improved packaging in vehicle interiors. • **High design freedom**, allowing seamless integration into complex, curved or segmented surfaces and enabling new interior design concepts. • **Quasi-invisible integration**, improving perceived interior quality and freeing visible space for other functions or materials. • **Enhanced functional flexibility** through array processing (adaptive directivity, zonal sound, tailored sound fields). This combination of **ultra-thin form factor, design flexibility and advanced acoustic control** opens new options for OEMs and system suppliers to differentiate their future mobility interior concepts.
Rationale for Joint R&D Project	For more than 20 years, Fraunhofer IDMT has conducted leading research in 3D audio and flat loudspeaker technologies, working closely with companies in automotive, professional, and high-end audio. These partners translate market observations (e.g. demand for new in-car sound experiences) into new features and product ideas. Many of them are SMEs acting as technology suppliers to major automotive manufacturers and facing strong cost, quality, and innovation pressure. A joint R&D project between a Korean SME and Fraunhofer IDMT offers clear advantages: • **Access to world-class expertise without high upfront investment** The SME can rely on Fraunhofer IDMT's long-standing know-how and infrastructure in 3D audio and loudspeaker design instead of building all competencies in-house (1), (2), (3), (4), (5). • **Cost-efficient and low-risk development** Joint R&D with an experienced research partner is often more economical and less risky than fully internal development, especially for advanced audio concepts. • **Structured translation of market needs into technical requirements** Fraunhofer IDMT supports the SME's acoustic team in converting feature and product ideas into a precise technical parameter list, ensuring clear targets for development and later industrialization. • **Benchmarking and technology evaluation** Fraunhofer helps analyze and evaluate state-of-the-art solutions, positioning the SME's products against global competitors and identifying unique selling points. • **Tailor-made technology for automotive audio** Fraunhofer IDMT offers development support for innovative loudspeaker solutions and 3D audio reproduction concepts that can be adapted to specific OEM requirements in Korea and worldwide. Korea, with its global top tier- consumer electronics companies and its position as the **third largest- electronics producer worldwide**, offers an excellent industrial environment to transform advanced research into competitive products. In this cooperation model, Fraunhofer IDMT develops and validates key technologies, the Korean SME integrates these results into its products, and

	automotive OEMs receive differentiated, high quality audio solutions. This -three-party constellation (OEM – Korean SME – Fraunhofer) can simplify project execution, reduce technical risk, shorten time- to- -market, and strengthen the SME's position as an innovative supplier in these mentioned markets (e.g. automotive audio, communication systems for trains, airplanes, ships).
Key Words	1. flat loudspeaker 2. integrated (invisible) sound system 3. audio system of the future 4. interior of the future

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 2	End of the Project	Level 6
R&D Objectives	The objective of this project is to develop a flat loudspeaker technology for a more space saving audio system installations. This might cover different interieur concepts of the future for cars, trains, airplanes, ships. Specifically, the project aims to: *Objective 1.* **Translate market and product requirements into technical specifications** Define clear acoustic, mechanical, electrical, and integration parameters for flat loudspeaker systems. *Objective 2.* **Design an innovative flat loudspeaker concept** Develop a loudspeaker architecture that meets the defined technical specifications and is suitable for various transportation interiors. *Objective 3.* **Develop and build a functional technology demonstrator** Implement the concept in a prototype system, including the necessary electronics and signal processing (where applicable). *Objective 4.* **Integrate and validate the demonstrator in an application-like environment** Test the prototype under realistic installation and operating conditions (e.g. in a vehicle mock-up) and verify performance targets. *Objective 5.* **Derive design guidelines for future flat loudspeaker products** Formulate practical guidelines and best practices for SMEs and OEMs to design, adapt, and industrialize loudspeakers based on the developed concept.			
R&D Target Specifications	The target performance specifications of the joint R&D project are defined to create a clearly differentiating, flat loudspeaker solution that can be industrialized for automotive and other mobility applications: 1. **Ultra-slim form factor for flexible integration** Reduce loudspeaker installation depth **to ≤ 20 mm**, enabling integration into thin structural layers such as car roofs, dashboards, headrests, door panels, and similar surfaces in trains, airplanes and ships. 2. **Adaptability to complex, non-planar surfaces** Develop **speaker array concepts** that can conform to **curved and non-planar geometries** (e.g. rounded dashboards, A/B/C pillars), maximizing design freedom for interiors. 3. **Optimized placement for improved user experience** Enable loudspeaker placement in **acoustically and ergonomically optimal positions**, rather than being restricted by installation depth or geometry, to support advanced in-cabin sound scenarios (e.g. personalized zones, enhanced navigation prompts). 4. **High audio quality system approach** Achieve **sound quality comparable to current audio systems** fulfilling customer requirements for frequency response, SPL, distortion, and spatial sound reproduction. This combination of ultra-thin form factor, design flexibility and advanced acoustic control opens up new options for OEMs and system suppliers to differentiate their future mobility interior concepts.			

R&D Plan	**Phase 1: Draft of a loudspeaker concept (TRL 2 → 3)** In Phase 1, the project focuses on deriving a clear product definition from customer and market requirements. Target applications (e.g. door panel, roof liner, seat back, dashboard) and boundary conditions (cost, packaging space, weight, acoustics, robustness) are collected and translated into a structured specification list. Several flat loudspeaker technology principles (e.g. micro-speaker arrays, panel / DML, hybrid concepts) are evaluated through theoretical considerations and preliminary simulations with respect to their ability to meet these specifications. On this basis, the most promising flat loudspeaker technology is selected. For at least one defined application scenario, an initial loudspeaker concept is drafted (including target geometry, array topology, acoustic target values). **Milestone Phase 1**: Selected flat loudspeaker principle and a documented concept specification (requirements + target architecture). **Phase 2: Develop a technology demonstrator (TRL 3 → 4)** In Phase 2, a technology demonstrator will be developed based on the loudspeaker concept defined in Phase 1. The demonstrator will comprise devices that already meet the relevant acoustic and geometrical specifications. To drive the loudspeaker demonstrator, laboratory electronic equipment will be used (e.g. DSP platforms, lab amplifiers). The demonstrator will be evaluated with respect to its basic functions in standardized test environments, for example in an anechoic chamber. The **milestone of this phase 2** is a fully functional loudspeaker demonstrator that is ready for adaptation to the defined application scenario (e.g. car cabin, roof module, seat back). **Phase 3: Integration and evaluation in the application-like area (TRL 4 → 5)** Phase 3 aims to integrate and evaluate the loudspeaker in an application-like area. For the evaluation the loudspeaker will be integrated and tuned for a certain sound application and tested by measurements and listening test. Beyond the fulfilment of the acoustic specifications, unwanted effects like noise caused by unsatisfying installations will be tested. Drawing on the test results, both the loudspeaker concept and its installation approach will be further optimized. The **milestone of phase 3** is a validated loudspeaker concept with it's potential for optimization. **Phase 4: Optimization for prototype state (TRL 5 → 6)** Phase 4 focuses on optimizing the concept based on the improvement potential identified in Phase 3. Depending on the availability of components representative of the final product (e.g. final loudspeaker hardware, electronic devices), requirements related to manufacturability and target cost ranges can also be addressed. This phase will result in guidelines for proper development and manufacturing of the loudspeaker system. The **milestone of phase 4** an optimized loudspeaker concept, including documented development and manufacturing guidelines

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IDMT) will lead the loudspeaker development. Its primary role is to suggest/develop a loudspeaker concept most promising to fulfil the acoustical and geometrical requirements. Specifically, the institute will: 1. Translation of the features into a list of technical specifications 2. Design and theoretical proof of a basic loudspeaker concept addressing these specifications 3. Evaluation of technical demonstrators regarding its acoustical performance compared to the specifications, deriving improvements if needed 4. Tuning and evaluation of the (improved) loudspeaker demonstrator integrated into an application-like area 5. Displaying optimization potential for the final state and deriving guidelines for speaker manufacturing and integration. In addition, Fraunhofer IDMT will provide research infrastructure, computing resources, and expertise in acoustic signal processing, and will be responsible for disseminating scientific outcomes through publications, patents, and technical documentation.
Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in real-world deployment, field validation, and commercialization-oriented development of the proposed technology. While the research institute focuses on core technology innovation, the SME will ensure that the developed methods are practical, scalable, and aligned with market requirements. Specifically, the SME partner will: 1. Provide feature list of the focused application scenario for the loudspeaker. 2. Build the technical demonstrators according to the specifications of the research institute. 3. Integrate the demonstrator into a self-provided application-like test set-up. 4. Rebuild the technical demonstrator according to optimization aspects. 5. Contribute to productization and commercialization planning, including integration into existing solutions and market validation. Through these roles, the SME partner will ensure that the proposed technology is not only scientifically sound but also economically viable and deployable at scale.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to generate both scientific progress and tangible industrial value. For the research institute, the project provides the opportunity to apply and further develop its expertise in advanced loudspeaker and audio system technologies on real-world, future-oriented mobility applications. This will lead to new scientific insights, demonstrators and publications, strengthen its international visibility in automotive and transport audio, and create a solid basis for further joint projects and IP. For the Korean SME, the collaboration enables access to cutting-edge flat loudspeaker concepts, signal processing methods and integration know-how that would be difficult to develop alone. The resulting technology will allow the SME to offer competitive, differentiated loudspeaker solutions for modern sound systems and interior concepts in cars, trains, airplanes and ships. This can improve its market position as an innovative supplier, open up new customer segments, and support export activities. By combining the institute's research excellence with the SME's industrial capabilities and market access, the project increases the likelihood of a successful transfer of the developed solutions into commercial products.

6. References

(1) Patent US2012008812 AA, Loudspeaker

(2) Patent US9854363 BB, Loudspeaker System

(3) Wave Field Synthesis Based Concept Car for High-Quality Automotive Sound, AES Conference on Automotive Audio, 2012, Munich, Germany

(4) Sladeczek et al: Object-Based audio as Platform Technology in Vehicles, AES Conference on Automotive Audio, 2022, Dearborn, MI, USA

(5) Sladeczek et al: Automotive Audio: Audio Objects as Platform Technology in Vehicles, magazine article in The Audio Voice, 2022, https://audioxpress.com/article/automotive-audio-audio-objects-as-platform-technology-in-vehicles

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization